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Evaluation of Subjective Tinnitus Severity and Distortion Product Otoacoustic Emissions and Extended High-Frequency Audiometry

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Abstract

Objective. Tinnitus is a multifactorial phenomenon with quality-of-life detriments for those affected by it. We aim to establish a relationship between subjective tinnitus severity with objective audiometric data in the extended high frequency (EHF) from 9 to 16 khz and with distortion product otoacoustic emissions (DPOAE). We hypothesize that severe subjective tinnitus as measured by the Tinnitus Handicap Inventory (THI) does not correlate with increased hearing thresholds in the EHF range.

Study Design. Prospective.

Setting. Single Tertiary Care Center.

Methods. Patients identified with tinnitus and normal hearing thresholds within standard frequency range (250-8000 Hz) were consented for participation. Those with underlying otologic disease, trauma, radiotherapy, or ototoxic drug use were excluded. The THI questionnaire was given to eligible patients and audiometric test results were collected. THI scores were categorized by severity groups. An n = 20 to 30 was determined to have an effect size of 0.7 with a significance level of $P = .05$.

Results. THI and audiometric data were collected for 38 patients and categorized into mild (n = 18, 47.4%), moderate (n = 8, 21.1%), slight (n = 7, 18.4%), and severe (n = 5, 13.2%) tinnitus severity groups. Mean THI score was 32.3 ± 19.6 with a statistically significant difference in scores by assigned THI severity group ($P < .01$). There were no significant differences or linear relationship among hearing thresholds in EHF range or DPOAE stratified by subjective tinnitus group ($P = .49$, $r^2 = 0.10$)

Conclusion. Subjective tinnitus severity is not predictive of audiometric outcomes. This finding can be used as a counseling tool to help tinnitus patients manage symptoms, expectations, and overall treatment outcomes.

Keywords

extended high frequency, Tinnitus

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Tinnitus, a multifaceted condition, affects approximately 10% to 15% of adults.[1,2] Despite various hypotheses, its exact cause remains elusive. Studies have revealed that individuals with tinnitus exhibit poorer extended high-frequency (EHF) hearing thresholds and lower levels of distortion product otoacoustic emissions (DPOAE) compared to those without tinnitus, suggesting potential cochlear involvement in the condition's genesis.[3-8]

The perception of tinnitus varies greatly among individuals, posing a challenge for clinicians when counseling patients. While some studies have attempted to establish a link between subjective tinnitus assessments and audiometric findings in the EHF range, the results are inconsistent. For instance, 1 study found that tinnitus patients with hearing loss in the extended range tended to be older and reported higher scores on tinnitus questionnaires than those with normal extended-range hearing.[9] Conversely, a more recent investigation discovered no significant difference in Tinnitus Handicap Inventory (THI) scores between individuals with normal and abnormal EHF responses.[4] Notably, no studies have explored the relationship between subjective tinnitus severity and DPOAE, representing a gap in our current knowledge.

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Psychometric tools such as the THI are employed in clinical practice to quantify tinnitus severity and assess its impact on patients. However, relating tinnitus severity to audiometric findings has proven challenging for clinicians. While some studies have linked tinnitus severity with psychological aspects such as stress and depression, they have not adequately addressed the association between audiometric thresholds and psychological factors.[10]

In this study, we aim to investigate the correlation between subjective tinnitus severity, objective audiometric data in the EHF range (9-16 kHz), and DPOAE. Our hypothesis posits that severe subjective tinnitus, as measured by the THI, will not be associated with elevated hearing thresholds in the EHF range, shedding light on the intricate relationship between tinnitus perception and audiometric outcomes.

Methods

Study Design and Participants

This observational study was approved by the Jefferson Institutional Review Board (IRB) and conducted at multiple subsites of a single institution. Participants were enrolled if they met the following criteria: thresholds greater than 25 dB from 250 to 8000 Hz audiogram, age over 18, presence of tinnitus without apparent causes such as central nervous system (CNS) pathology or ototoxic drug usage, and absence of abnormalities or conductive pathology during otologic examinations. Exclusion criteria encompassed a history of middle ear disease, vestibular disorders, ongoing ototoxic drug or head and neck radiotherapy, head or ear trauma, underlying systemic conditions like hypertension or diabetes, or any previous otologic diseases.

Data Collection

Eligible participants provided informed consent and underwent comprehensive audiometric evaluations, which included conventional pure-tone audiometry as well as testing in the EHF range. Additionally, DPOAE testing was administered. To assess the impact of tinnitus, subjects completed the THI questionnaire, contributing to psychometric data collection.

Statistical Analysis

A sample size range of $n = 20$ to 30 was determined to ensure an effect size of 0.7 at a significance level of $P = .05$. Statistical analysis included Spearman's rank test to assess correlations between THI scores and audiometric results in both conventional and EHF ranges, as well as DPOAE. Categorical variables were compared using Chi-square and Fisher's exact test. Descriptive statistics were employed to report the relationships between tinnitus severity, audiometric findings, sociodemographic characteristics, and medical comorbidities, including *t*-tests and analysis of variance where appropriate.

Results

THI and audiometric data were collected for 38 patients and categorized into mild (n = 18, 47.4%), moderate (n = 8, 21.1%), slight (n = 7, 18.4%), and severe (n = 5, 13.2%) tinnitus based on THI scores. Mean THI score was 32.3 ± 19.6 with a statistically significant difference in scores by assigned THI severity group ($P < .01$). Demographic factors including age, sex, race, and smoking status did not differ significantly among tinnitus severity groups (**Table 1**). THI questionnaire breakdown revealed statistically

Table 1. Subject Demographics by THI Group

Category	N = 38	THI-mild N = 18 (47.4%)	THI-moderate N = 8 (21.1%)	THI-severe N = 5 (13.2%)	THI-slight N = 7 (18.4%)	Significance (by THI)
AGE. mean ± SD	38.8 ± 11.2	37.6 ± 12.7	43.8 ± 12.2	37.4 ± 8.0	36.7 ± 8.6	0.167 (moderate vs mild) 0.678 (severe vs mild) 0.841 (slight vs mild) 0.304 (severe vs moderate) 0.202 (slight vs moderate) 0.806 (slight vs Severe)
Gender						0.804
Female	25 (69.4%)	12 (75.0%)	6 (75.0%)	3 (60.0%)	4 (57.1%)	
Male	11 (30.6%)	4 (25.0%)	2 (25.0%)	2 (40.0%)	3 (42.9%)	
Race						0.876
Black/African	3 (8.3%)	1 (6.2%)	1 (12.5%)	1 (20.0%)	-	
American	3 (8.3%)	2 (12.5%)	1 (12.5%)	-	-	
Other White	30 (83.3%)	13 (81.2%)	6 (75.0%)	4 (80.0%)	7 (100.0%)	
Smoking						
Current smoker	2 (5.6%)	1 (6.2%)	-	1 (20.0%)	-	0.062
Former smoker (quit)	8 (22.2%)	3 (18.8%)	2 (25.0%)	3 (60.0%)	-	Fisher's Exact
Never smoker	26 (72.2%)	12 (75.0%)	6 (75.0%)	1 (20.0%)	7 (100.0%)	

significant differences in all questions (except question 2: "Does the loudness of your tinnitus make it difficult for you to hear people?") by severity group (Supplemental Table S1, available online).

There were no statistically significant differences among severity groups across AC and DPOAE groups (Supplemental Table S2, available online). Our analysis, however, revealed frequency-specific variations in hearing thresholds among different tinnitus severity groups. Some frequencies, such as 500 and 1000 Hz, showed more pronounced differences between severity groups, while others exhibited fewer significant variations.

Additionally, no significant differences or linear relationship among hearing thresholds in EHF range or DPOAE stratified by subjective tinnitus group ($P = .49$, $r^2 = 0.10$) were noted.

Discussion

Tinnitus is a challenging clinical condition characterized by the subjective perception of ringing or buzzing sounds in the ears. Variability in its presentation poses difficulties for clinicians when counseling patients.[11] To address this issue, psychometric tinnitus questionnaires like the THI are employed to quantify tinnitus severity.[12] Limited research, however, has explored the relationship between subjective and objective measures of tinnitus. Our prospective cohort study aimed to establish a correlation between these measures, providing valuable guidance for clinicians and patients in managing tinnitus, a complex condition.

Our study revealed significant differences in THI scores among severity groups, indicating that patients' subjective experiences of tinnitus vary widely.[13] This variance extended to the impact of tinnitus on mental health, interpersonal relationships, and overall quality of life, highlighting the psychosocial strain tinnitus imposes.[14] These findings emphasize the importance of clinicians being attuned to the psychological aspects of tinnitus, as awareness of these issues can foster better clinician-patient relationships. Additionally, our results align with prior research suggesting a connection between tinnitus severity and psychosocial distress.[10,15] Notably, our study contributes to the current understanding of tinnitus by studying the correlation between audiometric thresholds and THI.

Aligned with our expectations, the current study did not find a significant correlation between subjective tinnitus severity, as measured by THI, and objective clinical severity indicators, such as hearing thresholds in the EHF range or DPOAE. This suggests that patients' self-reported tinnitus severity may not be a reliable clinical indicator for treatment guidance. These findings echo a common theme in the literature: tinnitus perception and effects vary widely between individuals, and there may not be a clear link between subjective and objective measures.[13,16] Our study, distinguished by its analysis of DPOAE thresholds and larger sample size, strengthens these observations. Based on these findings, patients and providers can feel optimistic about tinnitus treatment options and outcomes regardless of hearing loss severity.

Limitations

This study has several limitations that should be acknowledged. First, the reliance on the THI questionnaire introduces potential subjectivity and response bias. Despite efforts to increase the sample size, the stratification into tinnitus severity groups may not capture the full spectrum of tinnitus experiences, reducing generalizability of our results. The study primarily focuses on overall relationships, overlooking potential variations within subgroups of tinnitus patients. The use of conventional audiometry and DPOAE as assessment tools may not comprehensively reflect the complexity of tinnitus. The prospective cohort study design provides a static snapshot, missing the dynamic nature of tinnitus. Strict exclusion criteria may limit the study's relevance to real-world clinical populations, and the correlation analysis does not establish causation. These limitations highlight the need for future research to address these issues, explore alternative assessment methods, and consider subpopulations to enhance our understanding of tinnitus.

Conclusion

Our study underscores the challenges of quantifying tinnitus severity and highlights the disconnect between subjective and objective measures. Clinicians should recognize the substantial psychosocial burden of tinnitus on patients and maintain open communication to address patient experiences. While subjective measures may not provide precise clinical guidance, they remain crucial for understanding the multifaceted impact of tinnitus on patients' lives. Further research is warranted to explore alternative assessment tools and subgroups of patients to refine our understanding of tinnitus.

Author Contributions

Sruti Tekumalla, conceptualization, methodology, data acquisition and analysis, writing, review, and editing; **Natalie M. Perlov**, conceptualization, methodology, data acquisition and analysis, writing, review, and editing; **Saket Gokhale**, data acquisition; **Samiat Awosanya**, conceptualization, methodology; **Zachary D. Urdang**, review, editing; **Julia Croce**, methodology, review, editing; **Anna Bixler**, methodology, review, editing; **Thomas O. Willcox**, supervision, review, editing; **Rebecca C. Chiffer**, supervision, review, editing; **Dennis Fitzgerald**, supervision, conceptualization, methodology, review, editing.

Disclosures

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Supplemental Material

Additional supporting information is available in the online version of the article.

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